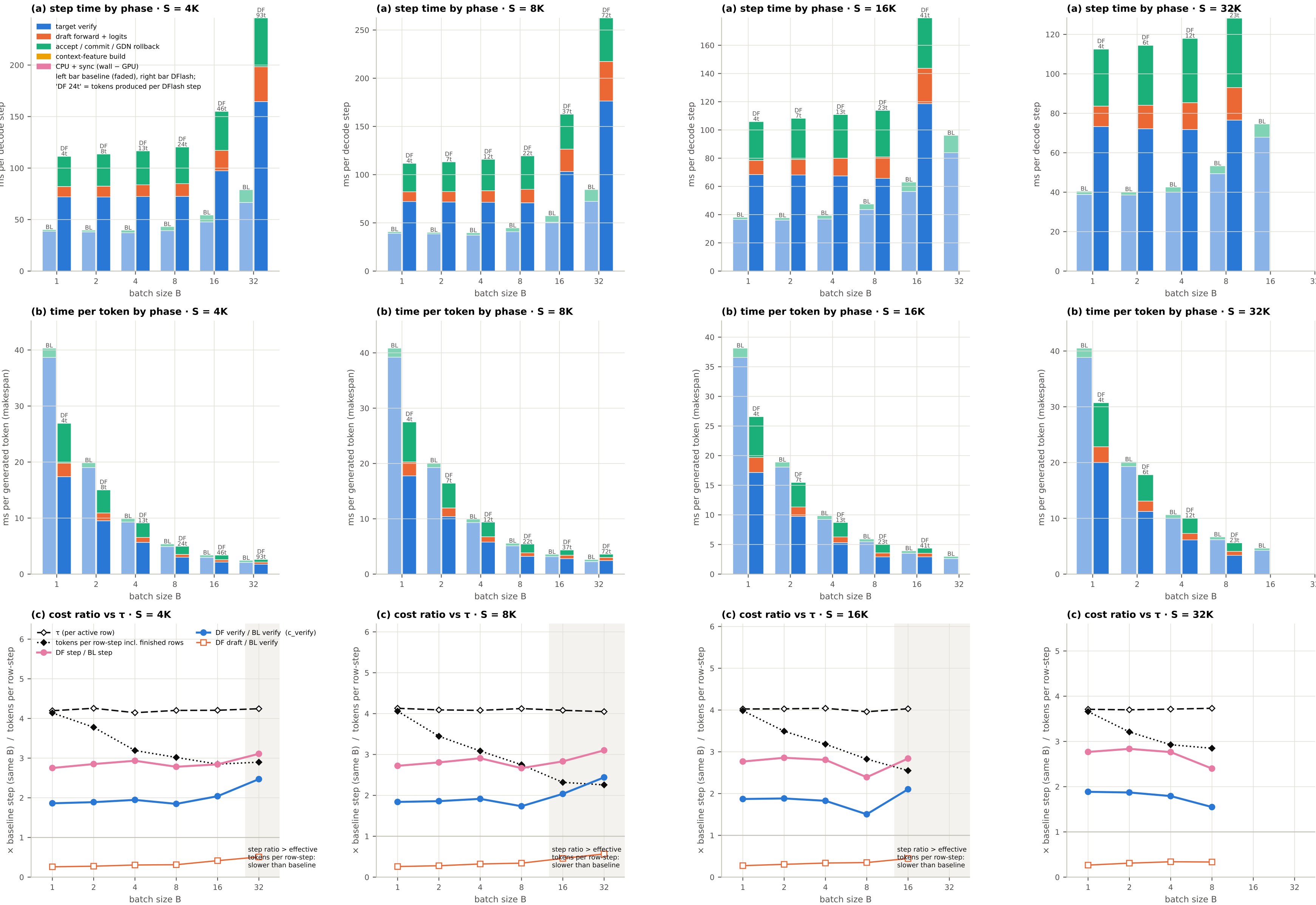


Qwen3.5-9B — where a decode step's time goes as B grows



GPU phases are CUDA-event means per step; CPU + sync is wall - Σ GPU (can be slightly negative, drawn as 0). (c) Makespan speedup = (tokens per row-step incl. finished rows) / (DF step / BL step) exactly — the magenta line crossing the dotted line is the break-even. The gap between the dashed τ and the dotted line is the static-batch tail (finished rows still verified). c_verify near 1 means 15 extra tokens per row are nearly free (memory-bound); it rises once the verify is compute-bound. 9B, torch fallback (no fla): accept/commit is the GDN rollback (replaying the kept prefix through the chunk kernel) for DFlash, and for the baseline a torch.where over every recurrent state it does not need — a baseline cost that grows with B and flatters DFlash at large B.